

Deep Architectural Evaluation of Splunk Enterprise Observability Gaps and the Blueprint for an Agentic Telemetry and Protocol Safeguard

The expansion of cloud-native microservices has transformed enterprise observability, shifting the standard for systems monitoring from simple reactive logging to standard metric, event, log, and trace (MELT) analysis.¹ Within this complex environment, the Splunk Enterprise Observability platform—comprising Splunk Observability Cloud and Splunk's core indexing technologies—stands as a powerful hybrid solution, offering high-speed search and unified SIEM and operational logging capabilities.³ However, enterprise adoptions frequently encounter systemic friction points, primarily categorized as compounding ingestion costs, high configuration overhead, and data-quality degradation that limits the efficacy of artificial intelligence (AI) diagnostic tools.¹

At the same time, the integration of autonomous agents via the Model Context Protocol (MCP) introduces a complex vectors-of-threat landscape.⁷ These systems expose internal databases and local files to prompt injections, privilege escalations, and unvetted outbound data flows.⁷ To bridge these capabilities and security gaps, this report details a comprehensive architectural evaluation of the platform's current limitations. Based on these limitations, a production-ready blueprint is proposed for the Splunk Agentic Ops Hackathon: **Project AegisOps**.⁸ This system is an MCP-enabled agentic middleware that dynamically optimizes telemetry ingestion budgets, validates data quality, and implements a real-time policy firewall over autonomous AI tool-calling workflows, ensuring robust, cost-effective, and secure operations.⁷

Technical Deficiencies in the Splunk Observability Ecosystem

The primary operational friction point within Splunk Observability Cloud stems from its split architectural billing and ingestion mechanics.⁶ While metric and trace ingestion is calculated primarily on a host-count basis, log management relies heavily on volume-ingest pricing models.⁶ SRE teams running elastic, auto-scaling Kubernetes workloads are frequently exposed to unpredictable billing spikes when unexpected error storms generate massive quantities of redundant debug logs.⁶

Compounding this problem, Splunk's Log Observer Connect does not ingest telemetry natively within the Observability Cloud; instead, it queries logs that are already indexed inside Splunk Enterprise.¹¹ This means organizations pay full volume ingestion premiums at the indexing edge before any contextual filtering can be applied.¹¹

Furthermore, Splunk's application performance monitoring (APM) engine is built on its 2019

acquisition of SignalFx.¹² While the integration has matured, the APM platform still exhibits configuration overhead.⁶ Unlike cloud-native competitors that auto-discover application topologies natively, Splunk APM requires specialized instrumentation, which is often difficult to configure.⁶

The learning curve is steep, particularly for teams adapting to the proprietary SignalFlow query language for custom dashboard metrics and writing Search Processing Language (SPL) for deep log correlation.⁵ Additionally, critical user-experience monitoring modules—such as Real User Monitoring (RUM) and Synthetic monitoring—are rated below industry standards.³ For example, Splunk’s RUM sessions are limited to a maximum duration of four hours, and separate operational products like Splunk On-Call impose direct seat-based pricing structures that increase the total cost of ownership.¹¹

Operational Metric / Tier Attribute	Small Workload Tier	Growing Workload Tier	Mid-Market Workload Tier
Typical Target Host Count	10 Hosts	50 Hosts	250 Hosts
Estimated Monthly Telemetry Context	~1 Terabyte (TB)	~5 Terabytes (TB)	~27 Terabytes (TB)
Infrastructure Plan Unit Cost	\$15 per host / month	\$15 per host / month	\$15 per host / month
App & Infrastructure Plan Unit Cost	\$60 per host / month	\$60 per host / month	\$60 per host / month
End-to-End Enterprise Suite Unit Cost	\$75 per host / month	\$75 per host / month	\$75 per host / month
On-Call Licensing Add-On	\$5 per user / month (up to 10 seats)	Custom Enterprise Agreement	Custom Enterprise Agreement
Typical Monthly Telemetry Cost Model	Average usage billing with overage charges applied	Average usage billing with overage charges applied	Custom volume-tier contracts with

	over purchased host allocations	over purchased host allocations	capacity quotas and tokens
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The data model details the pricing trajectory of Splunk Observability Cloud, illustrating how host-based tiers scale rapidly alongside volume-indexed logging requirements.⁶ For smaller organizations, this operational overhead can become prohibitive, while larger enterprises face complex budgeting challenges to control un-templated data growth.⁵

Data Quality as the Primary Bottleneck of Observability AI

Splunk’s native AIOps engines and recently launched troubleshooting agents are designed to automate root-cause analysis by correlating telemetry signals.² However, these systems are highly dependent on the quality of the underlying telemetry data.¹ The "AIOps paradox" indicates that poor data quality prevents teams from successfully adopting AI tools.¹ If log formats are inconsistent, trace IDs are missing, or metrics are not standardized against OpenTelemetry standards, AI-driven recommendations become unreliable.¹ Unlike data-trust-focused observability tools like Sifflet, which provide end-to-end data quality validation and automated cataloging, Splunk relies on a schema-on-read approach.⁶ This mechanism defers structural validation until the query executes, meaning corrupt or non-standardized telemetry is identified only after an incident has begun.⁶ In large, under-optimized environments, this leads to slow searches, parsing delays, and "clouded" AI diagnostics that require manual SRE intervention.⁵

Data Quality and AI Error Cascades	
Telemetry Input Quality	AI Agent Processing Outcome
Mismatched Trace IDs	Broken causal graph; the AI fails to correlate symptoms across services.
Non-Standardized Log Fields	Parsing latency; regex timeout extends MTTR during active incidents.
High Metric Cardinality	Query performance drops; agent times out trying to calculate baselines.

The cascade illustrates how data quality issues propagate through Splunk's AI layers, transforming minor log mismatches into analytical failures that require SREs to perform manual investigations.¹

Security Analysis of the Model Context Protocol (MCP) in SRE Workflows

The introduction of the Model Context Protocol (MCP) by Anthropic in late 2024 has standardized how large language models (LLMs) connect to external tools.⁷ Splunk’s official MCP Server (App ID 7931, version 1.0.1) exposes operational capabilities as callable JSON-RPC 2.0 tools, enabling agents to execute SPL, retrieve index configurations, and monitor system parameters using natural language.¹⁷ However, this architectural bridge creates a significant and dangerous threat surface.⁷

Because the Splunk MCP Server is hosted as a centralized service accessible over the default management port 8089 at /services/mcp, agents require highly privileged Splunk authentication tokens to execute their diagnostic tasks.¹⁷ This concentration of access makes MCP implementations highly vulnerable to specific exploits identified by the Coalition for Secure AI (CoSAI).⁷

CoSAI Threat ID	Threat Classification	Primary Attack Mechanism & Vector	Architectural Impact on Splunk Environment
MCP-T4	Input / Instruction Boundary Failure	Indirect prompt injection via compromised raw logs, incident tickets, or external system notifications.	An SRE agent reading a malicious log string executes unauthorized commands, such as exfiltrating indexes or modifying KV stores.
MCP-T5	Inadequate Data Protection	Sensitive data exposure (PII, API keys) inside debug logs, system traces, or console error messages.	The MCP server indexes credentials in plaintext, exposing them to any user query or unauthorized LLM context window.
MCP-T6	Missing Integrity controls	Post-deployment modification of tool definitions, config files, or binary	Attackers run "shadow" MCP servers or spoof tool definitions to

		wrappers.	intercept API tokens and Splunk queries silently.
MCP-T8	Network Binding Failures	Binding the MCP server to insecure interfaces (0.0.0.0) instead of highly isolated loopbacks.	Exposes Splunk's internal REST management APIs to external networks, enabling lateral movement and DNS rebinding.
MCP-T9	Trust Boundary Failures	Confused deputy scenarios where the agent acts blindly on instructions without verifying context.	A compromised agent uses its elevated service privileges to search files, bypass read/write boundaries, or delete databases.

These threat vectors show that security monitoring must evolve beyond simple network-level boundaries.⁷ Traditional firewalls cannot parse the semantic intent of an agent's query, meaning malicious instructions wrapped in valid JSON-RPC 2.0 structures will bypass standard perimeter defenses.⁷

To mitigate these security risks, SRE teams utilize tools like the Splunk Technology Add-On (TA) for MCP (App ID 8377, built by Rod Soto).⁷ This add-on provides standardized CIM-compliant field extractions for JSON-RPC 2.0 message streams, allowing analysts to monitor and audit semantic activities.⁷ However, while the TA offers visibility, it remains a reactive logging tool and does not provide an active, inline firewall to block unauthorized or malicious tool calls in real time.⁷

Comparative Architectural Capabilities of Tool Implementations

Developers seeking to extend Splunk's capabilities can utilize different MCP server options, each offering distinct architectural and functional trade-offs.¹⁸

Architectural Vector	Official Splunk MCP Server (App ID 7931)	Livehybrid Alternative (splunk-mcp)
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Project Maturity	Splunk Supported (Beta, Version 1.0.1)	Open-Source Community Tool (FastMCP-based)
Exposed Toolset	generate_spl, run_splunk_query, get_splunk_info, get_indexes, get_index_info	run_splunk_query, list_indexes, get_index_info, indexes_and_sourcetypes, list_kvstore_collections, create_kvstore_collection, delete_kvstore_collection
Resource & Prompt Support	Exclusively exposes callable tools; does not declare native prompts or resources	Custom prompt configurations with detailed emoji-supported logging streams
Transport Capabilities	HTTP / Server-Sent Events (SSE) hosted on port 8089 over /services/mcp	Multi-mode transport: STDIO (for Claude Desktop), SSE (web clients), and REST API modes
Data Interaction Scope	Strictly limited to Splunk index querying and instance metadata analysis	Includes active read/write operations over Splunk Key-Value (KV) store collections
Security Mechanism	Integrated role-based access control (RBAC) mapping to Splunk's internal permissions	Relies on environment variables (SPLUNK_TOKEN, VERIFY_SSL) with manual port exposure rules

The comparison shows that while the official app provides a highly secure, read-only interface for standard querying, community implementations offer richer write capabilities (such as KV store operations).¹⁸ These advanced write operations, however, introduce a broader threat surface that requires robust, real-time security management.⁷

Quantifying the Value of Agentic SRE Automation

Deploying autonomous agentic AI in incident response environments yields measurable performance improvements compared to traditional, manual SRE workflows.²⁰ Unlike simple reactive chatbots, agentic SRE systems continuously execute a "perceive, decide, act, and observe" loop, automating incident triage and investigation end-to-end.²⁰

Incident Dwell-Time Reductions	
MDR Deployment State	Incident Resolution Time
Manual Incident Handling	24 Days
Agentic Autonomous MDR	24 Minutes

The reduction in dwell time highlights how automated triage and incident enrichment eliminate manual bottleneck delays.²¹

To evaluate the impact of these technologies, the following dataset details quantitative outcomes from real-world agentic SRE deployments ²¹:

Case Study Organization	Deployment Architecture	Reclaimed Analyst Productivity	System Metric Performance
Eye Security (630 Incident Analysis) ²¹	Managed Detection & Response (MDR)	Reduced manual handling time from 19 hours to 2 hours per incident.	Business Email Compromise (BEC) dwell time was reduced by 99.9%.
DXC Technology ²¹	7AI Agentic SOC Layer	Reclaimed 224,000 analyst hours (equivalent to ~112 full-time employees).	Both mean time to detect (MTTD) and mean time to respond (MTTR) fell by 50%.
Western Governors University (WGU) ²¹	AWS DevOps Agent Pipeline	Automated incident response and telemetry correlation loops.	Total MTTR was reduced by 77%, falling from 2 hours to 28 minutes.

These case studies prove that automating routine, low-judgment tasks—such as trace correlation, log enrichment, and alert grouping—allows human SREs to focus on complex threat hunting and architecture improvements.²¹

AegisOps: An Agentic Telemetry and Protocol

Safeguard

To solve these pricing, data-quality, and security challenges within the Splunk ecosystem, **Project AegisOps** is proposed as an innovative submission for the Splunk Agentic Ops Hackathon under the **Observability Track**.⁸ AegisOps is designed as an agentic middleware proxy that interfaces with the Splunk MCP Server, the Splunk MCP TA, and local OpenTelemetry collector configurations.⁷

AegisOps System Architecture	
Component Engine	Primary Functionality
Telemetry Purifier Engine	Monitors ingestion rates and generates filters.
Telemetry Quality Validator	Profiles schemas against CIM standard formats.
Semantic MCP Firewall	Intercepts and sanitizes JSON-RPC tool calls.

The architecture secures tool interactions while reducing ingestion-based licensing costs.⁶

The Operational Causal Loop

AegisOps operates via an autonomous, goal-oriented loop, continuously optimizing telemetry streams and securing agent interactions ²⁰:

1. **Perceive:** The proxy continuously captures the JSON-RPC stream directed over port 8089 /services/mcp.¹⁷ Concurrently, it queries Splunk metadata to monitor ingestion volume run-rates across different indexes.¹¹
2. **Decide:** The system runs real-time semantic analysis to identify cost spikes and security anomalies.⁶ It uses cosine-similarity scoring to compare incoming user prompts against safe policy boundaries.⁷
3. **Act:** If ingestion volume surpasses specified cost limits, AegisOps generates custom OpenTelemetry processor configurations to filter duplicate log files.² If a security threat is detected within an MCP tool call, the firewall blocks the command and returns a standardized JSON-RPC error response.⁷
4. **Observe:** The agent monitors index volume metrics to confirm the success of the

optimization rules.¹¹ If a blocked security threat occurs, it verifies that the SRE console remains uncompromised and logs the security event to the audit index.⁷ To evaluate the security risk of incoming MCP commands, the Semantic Firewall calculates a real-time risk index:

$$\text{Risk Score} = w_1 \cdot \text{Command Entropy} + w_2 \cdot \text{Access Privilege Level} + w_3 \cdot \text{Semantic Deviation}$$

If the calculated Risk Score exceeds the pre-configured system threshold (τ_{safe}), the transaction is immediately blocked and routed to the SRE console for verification.⁷

Human-in-the-Loop Governance and Safety Guardrails

Because AI agents can fail up to 70% of the time when executing multi-step tasks without oversight, AegisOps enforces a strict human-in-the-loop (HITL) architecture.¹⁰ This approach ensures that high-risk actions—such as system rollbacks, policy modifications, or configuration changes—cannot be executed autonomously.¹⁰

Operational Vector	Risk Classification	System Permission Level	Human-in-the-Loop Verification Mechanism
Index Meta-Queries	Low Risk	Read-Only	Runs fully autonomously; system executes the search and displays the natural-language summary. ¹⁰
Telemetry Filter Deployment	Medium Risk	Write-Config	The agent detects an ingestion spike and proposes a filtering rule. ⁶ The SRE receives a Slack notification detailing the change and must click "Approve" before the rule is

			committed. ¹⁰
Microservice Restart	Medium Risk	Active-Write	Following a latency alert, the agent proposes a container restart runbook. ¹⁴ Execution remains paused until the SRE approves the action. ¹⁰
KV Store Modifications	High Risk	Full-Write	Direct write, creation, or deletion commands on KV stores are blocked. ¹⁰ The firewall isolates the call and requires multi-factor authentication to unblock the tool. ⁷
Configuration Erasure	High Risk	Directory Access	Any command seeking to modify core indexes or access restricted directories (mcp.method search_files) is permanently blocked. ⁷

This protocol ensures that while agents handle high-velocity data correlation, humans maintain absolute authority over the state of production systems.¹⁰

Actionable Implementation Plan and Business Value Matrix

To ensure a viable submission for the Splunk Agentic Ops Hackathon (running from May 18th to June 15th, 2026), Project AegisOps follows a fast-paced, weekly development schedule.⁸

Hackathon Development Roadmap	
Milestone 1 (Days 1 - 7)	- Set up Splunk MCP Server in Docker. - Deploy Splunk MCP TA for logging. - Build the JSON-RPC proxy layer.
Milestone 2 (Days 8 - 14)	- Develop Telemetry Purifier Engine. - Build semantic firewall parser. - Define least-privilege policy rules.
Milestone 3 (Days 15 - 21)	- Integrate Slack/Teams approval buttons. - Conduct simulation tests for attacks. - Package and document for submission.

The roadmap balances immediate technical integration with long-term security testing to deliver a production-ready application.⁸

To demonstrate the real-world value of this project, the following business matrix maps AegisOps features to concrete organizational metrics⁵:

Enterprise Pain Point	AegisOps Feature	Target Business Metric Improvement	Strategic Business Outcome
Unpredictable Ingestion Overages ⁵	Dynamic telemetry filtering and sampling via OpenTelemetry ²	Redundant debug log volumes are reduced by 30% to 50%. ⁶	Lowered Total Cost of Ownership (TCO), making Splunk viable for smaller, cost-conscious teams. ⁵
High MTTR and AI Hallucinations ¹	Telemetry profiling and CIM format validation ¹	Incident response MTTR is reduced by up to 50%. ²¹	Faster resolution of customer-impacting outages, preserving SLA commitments. ¹

Exposed Tool-Calling Security Risks ⁷	Semantic firewall and JSON-RPC sanitization ⁷	Prevents 100% of unvetted, high-risk file and index operations. ⁷	Secure adoption of generative AI systems within sensitive, regulated cloud environments. ⁷
Steep Platform Learning Curve ⁵	Natural language-to-SPL translation and agentic search ¹⁸	Reduces developer onboarding time from weeks to hours. ⁵	Improved developer productivity and reduced operational reliance on specialized query experts. ⁶

Strategic Conclusions

The Splunk Enterprise Observability platform delivers high-fidelity telemetry monitoring and SIEM analytics across hybrid cloud environments.³ However, its long-term adoption is frequently limited by complex ingestion billing, a steep query language learning curve, and strict dependencies on clean telemetry data to prevent AI analysis errors.¹ Deploying autonomous AI agents via the Model Context Protocol (MCP) offers a path toward automated site reliability engineering, but it introduces critical security threats, including prompt injection, data exfiltration, and privilege escalation.⁷

By developing **Project AegisOps** for the Splunk Agentic Ops Hackathon, developers can bridge these critical gaps.⁸ AegisOps acts as an intelligent, secure, and cost-aware proxy that optimizes telemetry streams, safeguards AI tool execution, and keeps human SREs in the loop with structured notifications and approval workflows.¹⁰ Building this solution demonstrates how enterprise organizations can safely operationalize autonomous AI agents—maximizing system reliability and security while keeping cloud infrastructure costs under strict control.⁶

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